QUALIFICATION REQUIREMENTS REMOVED

SUPERSEDING
MIL-M-38510/471
15 October 1980

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS/SOS

1024 BIT RANDOM ACCESS MEMORY (RAM)

MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION.

This specification is approved for use by George C. Marshall Flight Center, National Aeronautics and Space Administration, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic, silicon on sapphire, static, silicon gate CMOS 1024 bit random access memory microcircuits. One product assurance class and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be as specified in MIL-M-38510, with the exception that the "JAN" or "J" certification shall not be used.

1.2.1 Device type. The device types shall be as follows:

Device type	Circuit
01	1024 word/1 bit static RAM
02	256 word/4 bits static RAM

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline, (see MIL-M-38510, appendix C)
W	D-7 (22 lead, 0.4" x 1"), dual-in-line package
E	D-2 (16 lead, 1/4" x 7/8"), dual-in-line package

1.3 Absolute maximum ratings:

Storage-temperature range	-65°C to +150°C
Operating temperature range (T_C)	-55°C to +125°C
Supply voltage range (V_{DD} - V_{SS})	-0.5 V dc to +11 V dc
Maximum power dissipation, P_D:	
T_C = -55°C to +100°C	500 mW
T_C = +100°C to +125°C	Derate linearly at 12 mW/°C to 200 mW
Device dissipation per output transistor:	
T_C = -55°C to +125°C	100 mW
Input voltage range, all inputs	V_{SS} -0.5 V dc to V_{DD} +0.5 V dc
Input current (each input)	5 mA
Lead temperature (soldering, 10 seconds)	+265°C
Maximum junction temperature (T_J)	+150°C
Thermal resistance, junction to case (θ_{JC})	0.0833 °C/W

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: George C. Marshall Space Flight Center, National Aeronautics and Space Administration, ATTN: EO2, Marshall Space Flight Center, AL 35812, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions:

Supply voltage range (V_{DD} - V_{SS})	4 V dc to 10 V dc
Input low (V_{IL}) voltage range	0 V dc to 1.5 V dc at V_{DD} = 5 V dc; 0 V dc to 3 V dc at V_{DD} = 10 V dc
Input high (V_{IH}) voltage range	3.5 V dc to 5 V dc at V_{DD} = 5 V dc; 7 V dc to 10 V dc at V_{DD} = 10 V dc
Case operating temperature range	-55°C to +125°C
Timing parameters	See 6.4.1 and figures 7, 8, and 9

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Terminal connections. Terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Block diagrams. The block diagrams shall be as specified on figure 3.

3.2.4 Bit address map. The bit address map shall be as specified on figures 4 and 5.

3.2.5 Case outlines. The case outlines shall be as specified in MIL-M-38510 and in 1.2.3 herein.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics shall be as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. Electrical test requirements shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ V_{SS} = 0 V, $-55°C \leq T_C \leq +125°C$ Unless otherwise specified		Device type	Limits Min	Limits Max	Unit
Positive clamping input to V_{DD}	$V_{IC(pos)}$	T_C = +25°C; V_{DD} = GND; V_{SS} = Open; I_{IN} = 1 mA; outputs open		01,02	---	3.0	V
Negative clamping input to V_{SS}	$V_{IC(neg)}$	T_C = +25°C; V_{DD} = Open; I_{IN} = -1 mA; V_{SS} = GND; outputs open		01,02	---	-3.0	V
Quiescent supply current	I_{DD}	V_{DD} = 11.0 V (see table III)		01,02	---	1,000	µA
	I_{SS}			01,02	---	-1,000	µA
High-level output voltage	V_{OH1}	V_{DD}=5 V	V_{IN} (see table III)	01	4.5	---	V
				02	4.6	---	V
	V_{OH2}	V_{DD}=10 V	I_{OH} (see table III)	01	9.1	---	V
				02	9.5	---	V
Low-level output voltage	V_{OL1}	V_{DD}=5 V	V_{IN} (see table III)	01,02	---	0.4	V
	V_{OL2}	V_{DD}=10 V	I_{OL} (see table III)	01,02	---	0.5	V
High-impedance output leakage current (3 state)	I_Z	V_{DD} = 11 V;		01	---	±20	µA
		V_{IN} (see table III)		02	---	±20	µA
High-level input current	I_{IH}	V_{DD} = 11 V; V_{IN} = 11 V		01,02	---	10	µA
Low-level input current	I_{IL}	V_{DD} = 11 V; V_{IN} = 0 V		01,02	---	-10	µA
V_{DD} supply current low	I_{DDL}	V_{DD} = 5 V; V_{IN} = 0.0 V		01,02	---	500	µA
Operating supply current	I_{DDOP}	See table III		01	---	8	mA
				02	---	16	mA
Output valid time	TQVQZ	See table III	V_{DD} = 5 V	01	---	60	ns
			V_{DD} = 10 V	01	---	30	ns
Output enable time from $\overline{CS1}$	TSLQV	See table III	V_{DD} = 5 V	01	---	100	ns
			V_{DD} = 10 V	01	---	80	ns
			V_{DD} = 5 V	02	---	560	ns
			V_{DD} = 10 V	02	---	280	ns
Output disable time from $\overline{CS1}$	TSHQZ	See table III	V_{DD} = 5 V	01	---	85	ns
			V_{DD} = 10 V	01	---	55	ns
			V_{DD} = 5 V	02	30	---	ns
			V_{DD} = 10 V	02	30	---	ns
Read access time	TAVQV	See table III	V_{DD} = 5 V	01	---	320	ns
			V_{DD} = 10 V	01	---	200	ns
			V_{DD} = 5 V	02	---	630	ns
			V_{DD} = 10 V	02	---	315	ns
Output active from output disable	TGLQV	See table III	V_{DD} = 5 V	02	---	250	ns
			V_{DD} = 10 V	02	---	140	ns
Output hold from $\overline{CS1}$	TSHQV	See table III	V_{DD} = 5 V	02	30	---	ns
			V_{DD} = 10 V	02	30	---	ns
Output hold from CS2	TSLQZ	See table III	V_{DD} = 5 V	02	30	---	ns
			V_{DD} = 10 V	02	30	---	ns
Output hold from output disable	TGHQZ	See table III	V_{DD} = 5 V	02	30	---	ns
			V_{DD} = 10 V	02	30	---	ns
Power down	PWRDWN	See table III		01,02	---	2.0	V

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TABLE II. Electrical test requirements for device classification.

Line No.	Applicable tests and MIL-STD-883 test methods	Class B device 1/		
		Ref. par.	Table 2/ III sub-groups	Table 3/ IV delta limits
1	Interim electrical parameters method 5004		1, 7	
2	Static burn-in I method 1015			
3	Same as line 1			
4	Static burn-in II method 1015			
5	Same as line 1 2/			Δ
6	Dynamic burn-in method 1015	4.2b	4/	
7	Same as line 1 2/	4.2d	1*	
8	Final electrical parameters 2/ method 5004		1*, 2, 7*, 8	
9	Group A requirements method 5005	4.4.1	1,2,7, 8,9,10	
10	Group B end point electrical parameters method 5005			
11	Group C end points method 5005	4.4.3	1,2,7,8	Δ
12	Additional group C tests method 5005	4.4.3c		
13	Group D end points method 5005	4.4.4	1,2,7,8	

1/ Blank spaces indicate tests are not applicable.

2/ (*) indicates PDA applies to subgroup 1 and 7 (see 4.2e).

3/ (Δ) indicates delta limit shall be required and delta values shall be computed with reference to the previous interim electrical parameters.

4/ The device manufacturer may at his option either perform delta measurements or within 24 hours after burn-in (or removal of bias) perform the final electrical parameter measurements.

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3.5.1 Functional tests. The functional tests used to test these devices are contained in table III. If the test patterns cannot be implemented due to the test equipment limitations, then alternate test patterns to accomplish the same results shall be submitted to the contracting activity for approval.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container. The "JAN" or "J" certification mark shall not be used.

3.7 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have a manufacturer certification in accordance with MIL-M-38510 for at least one line; not necessarily the line producing the device type described herein.

3.8 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, as applicable, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following modifications and additional criteria shall apply:

a. Delete the sequence specified in 3.1.9 through 3.1.13 of method 5004 (MIL-STD-883) and substitute lines 1 through 7 of table II herein.

b. Burn-in test (method 1015 of MIL-STD-883).

(1) The burn-in test as described in 3.1.10 of method 5004 shall be performed as indicated. (3.1.11 and 3.1.12 are not required; 3.1.9 is required for class B devices.)

(2) For class B devices, dynamic burn-in shall be performed for 160 hours minimum at 125°C ambient temperature, using the circuit shown on figure 6, or equivalent.

c. Interim and final electrical parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

d. For class B devices, post dynamic burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter measurements.

e. The percent defective allowable (PDA) for class B devices shall be 10 percent based on failures from group A, subgroups 1 and 7 (after cooldown) of final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroups 1 and 7, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

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4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.6) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

A special subgroup shall be added using an LTPD of 15 for class B. This subgroup shall consist of a high-voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

4.4.3 Group C inspection. Group C inspection shall consist of the test subgroups and LTPD values shown in table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection (see table IV herein).

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E using the circuit shown on figure 6, or equivalent.

(2) T_A = +125°C, minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Additional electrical subgroups as specified in table II shall be added to the group C inspection requirements for class B devices.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883; end-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate table and as specified herein.

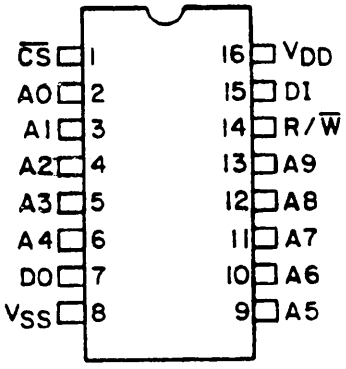
NOTE: In order to prevent "latch-up" during device turn-on and also prevent a high initial operating current (I_{DDOP}), set all device terminals equal to V_{SS}. Raise V_{DD} and $\overline{CS}$ simultaneously to normal system operating V_{DD}. Do not allow V_{IH} at $\overline{CS}$ to exceed V_{DD} +0.5 V while rising.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal, unless otherwise specified. Currents given are conventional current and positive when flowing into the referenced terminal.

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junction is defined in terms of a particular ordered pair of interface ... each primary

Device type 01



Device type 02

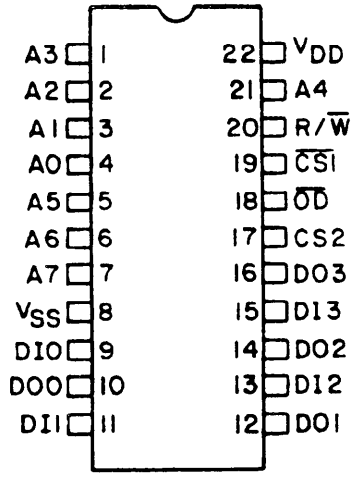


FIGURE 1. Terminal connections.

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Device type 01

MODE	INPUTS				OUTPUTS
	R/$\overline{W}$	CS	ADD	DI	DO
STANDBY	X	1	X	X	HI Z
WRITE "H" INTO CELL AXY	0	0	AXY	1	HI Z
WRITE "L" INTO CELL AXY	0	0	AXY	0	HI Z
READ DATA IN CELL AXY	1	0	AXY	X	(CONTENTS OF AXY)

Device type 02

FUNCTION	R/$\overline{W}$	$\overline{CS1}$	$\overline{CS2}$	$\overline{OD}$	DATA OUT DO
READ	1	0	1	0	Storage state of addressed cell
WRITE (OUTPUT DISABLED)	0	0	1	1	High impedance
WRITE	0	0	1	0	New data in state
STANDBY	X	1	X	X	High impedance
	X	X	0	X	High impedance
	1	0	1	1	High impedance

Logic 1 = High Logic 0 = Low X = Don't Care

FIGURE 2. Truth tables.

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Device type 01

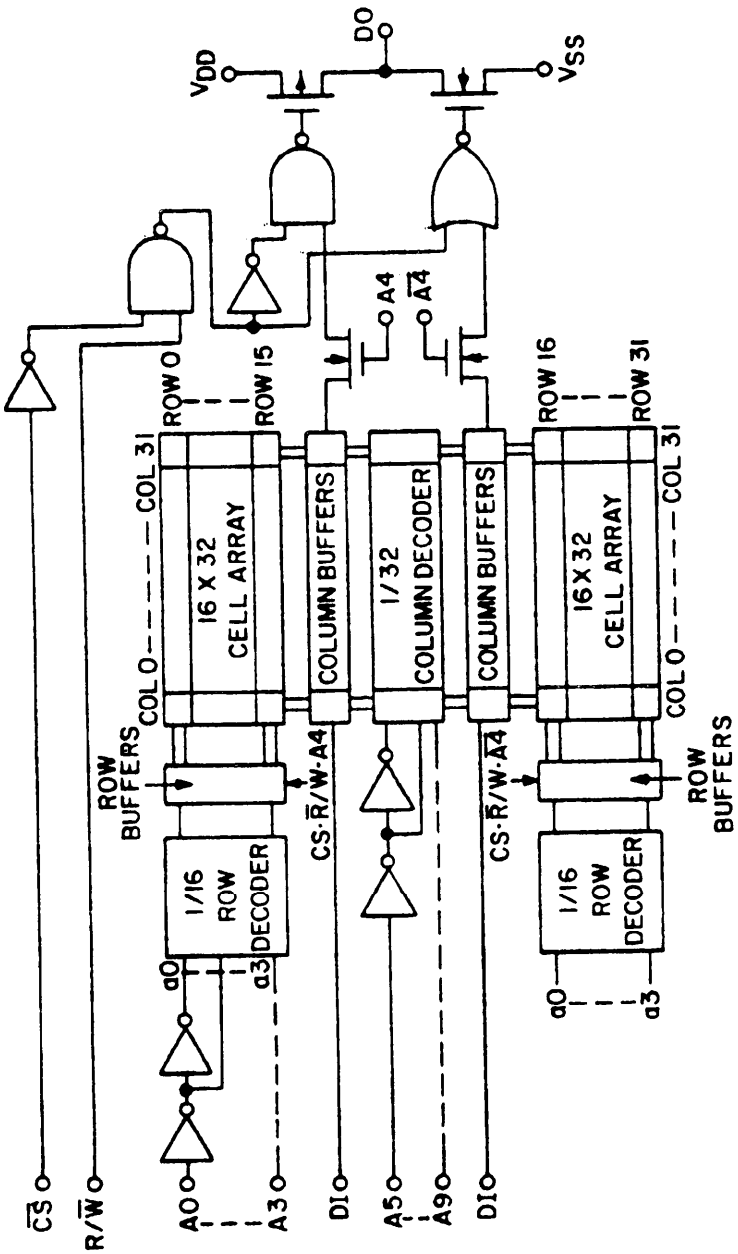


FIGURE 3. Block diagram.

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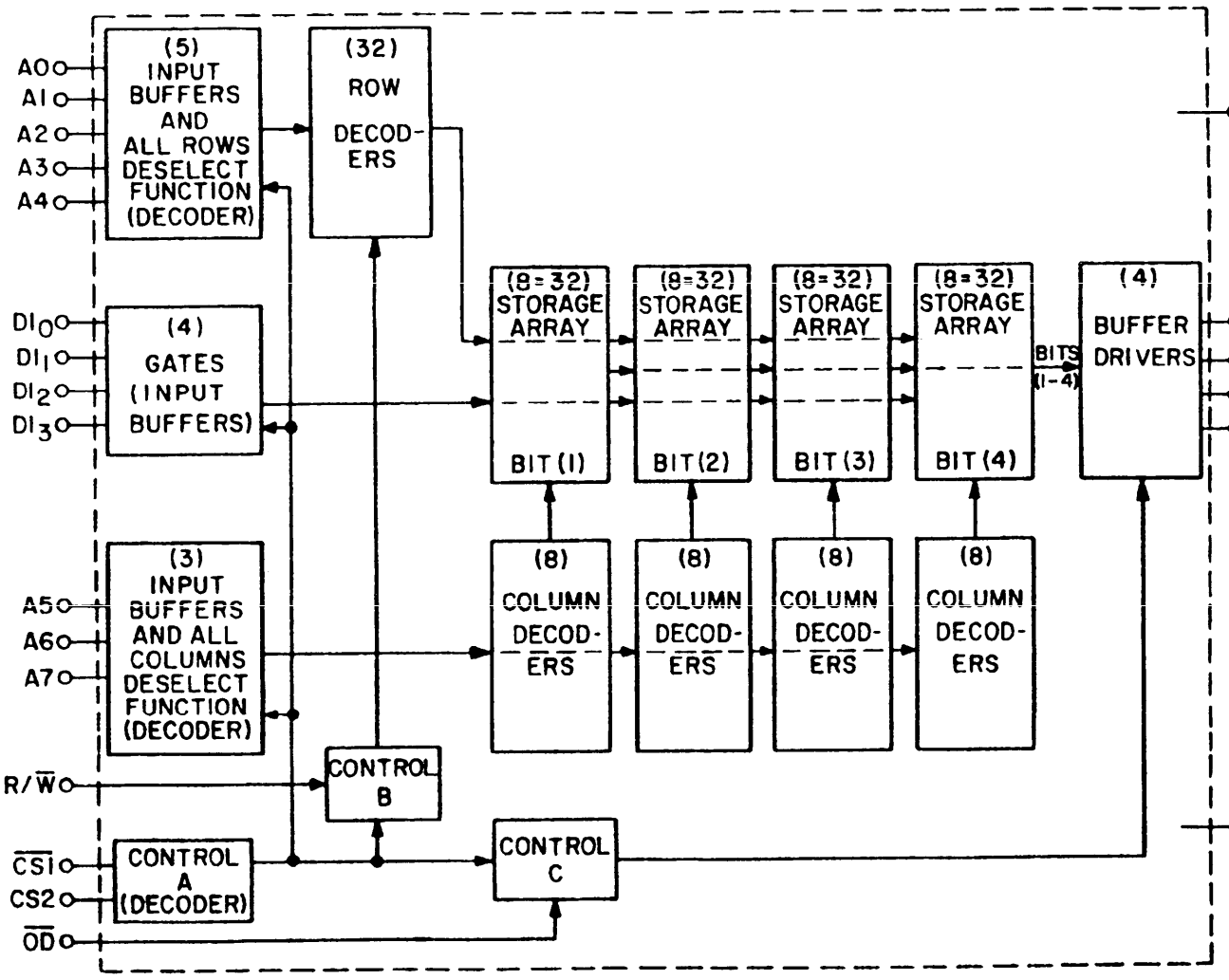


FIGURE 3. Block diagram – Continued.

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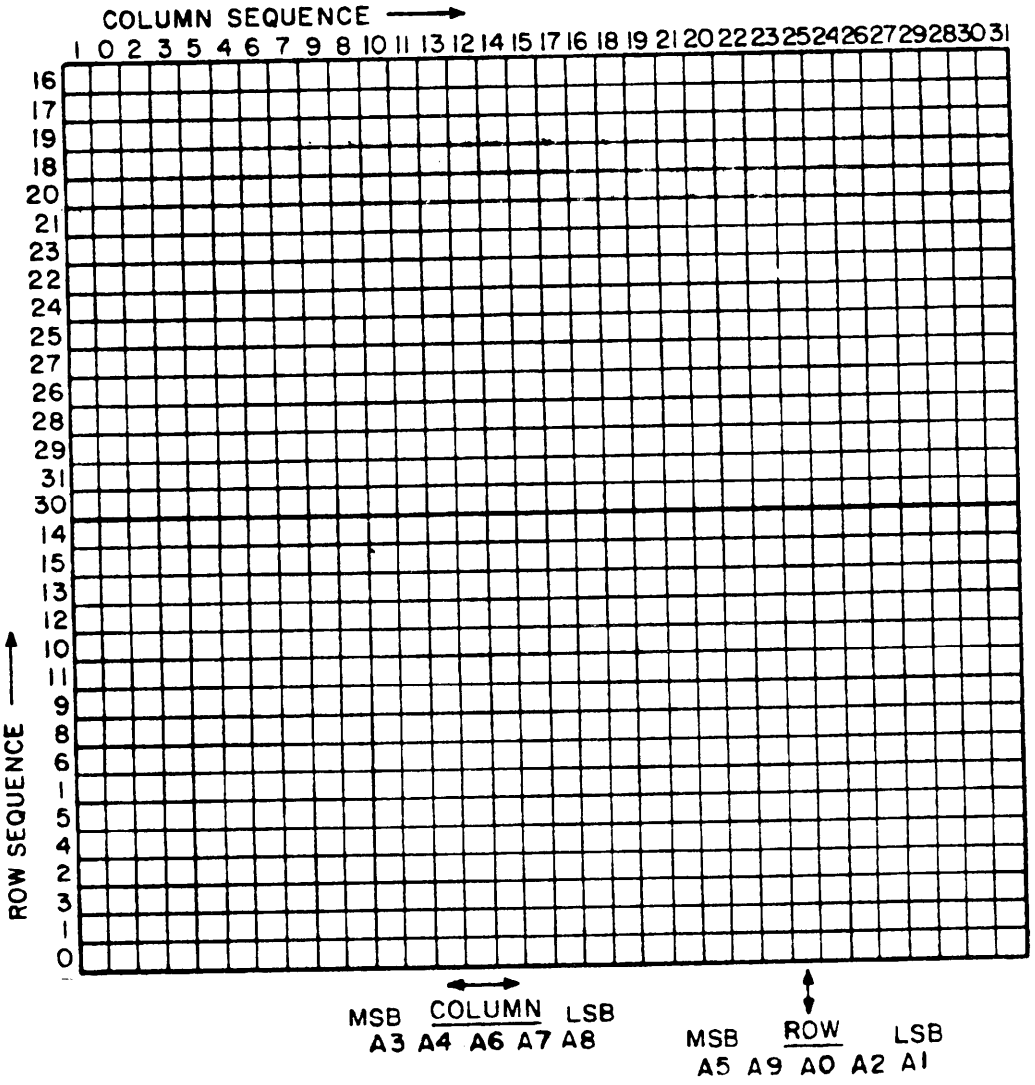


FIGURE 4. BIT map for device type 01.

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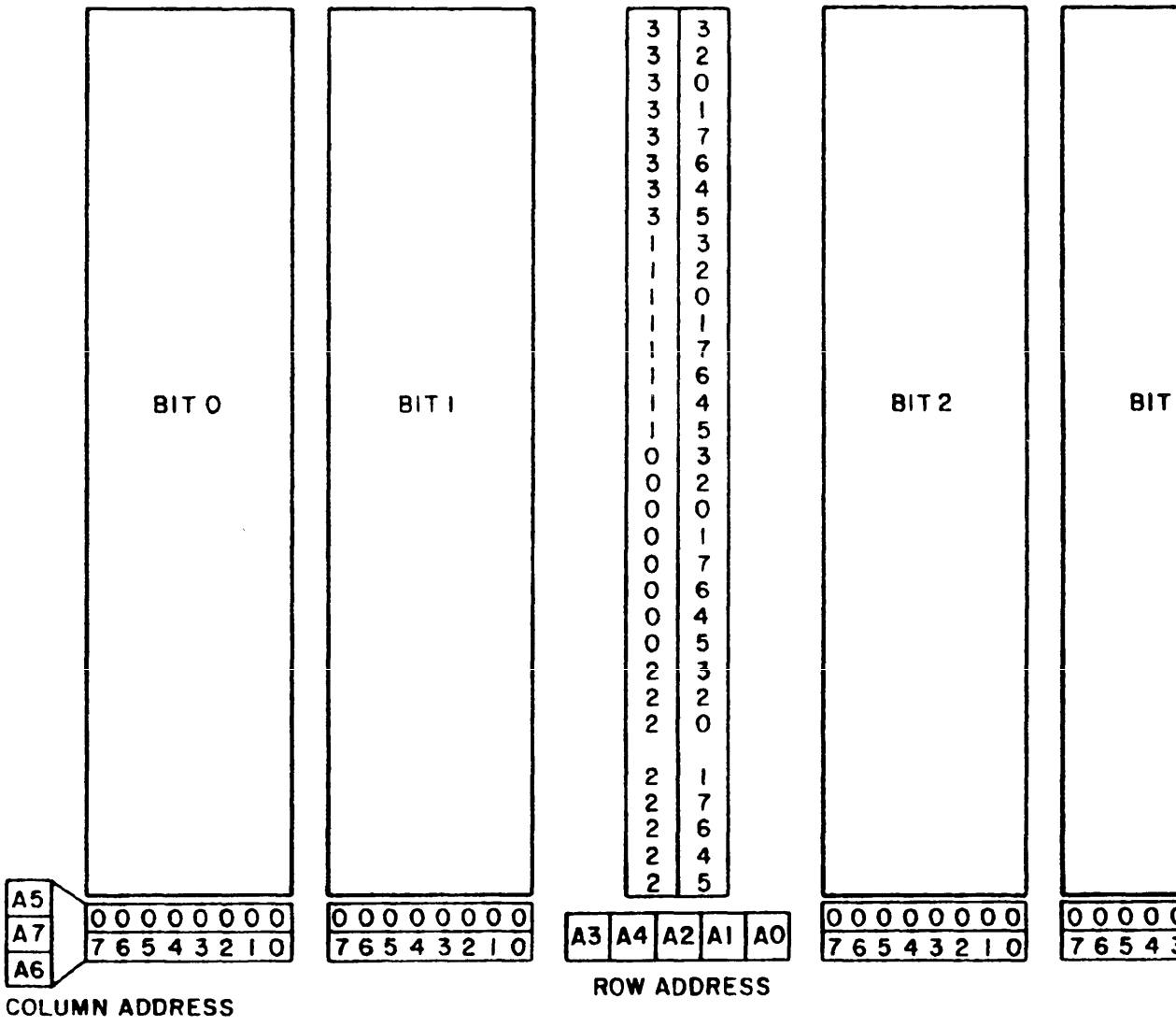
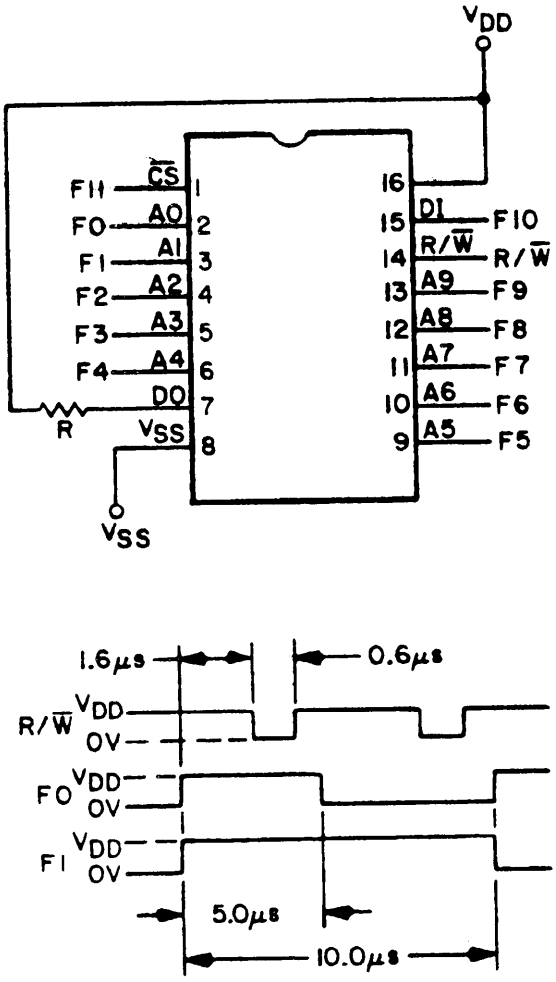


FIGURE 5. BIT address map for device type 02.

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Device type 01

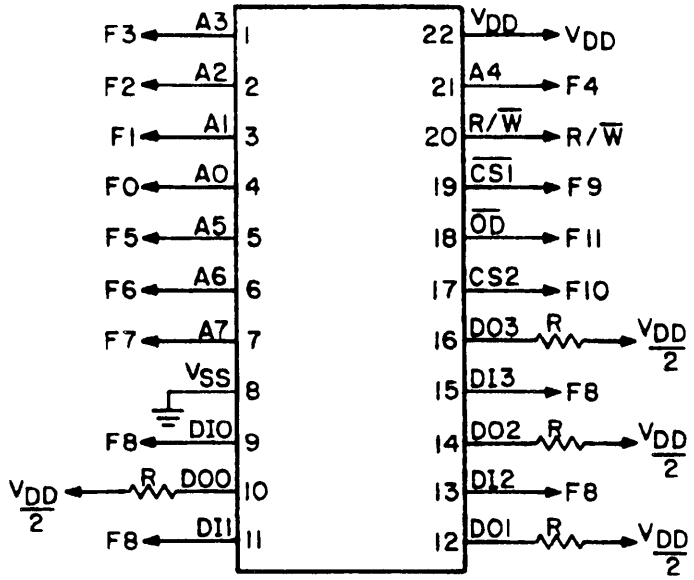


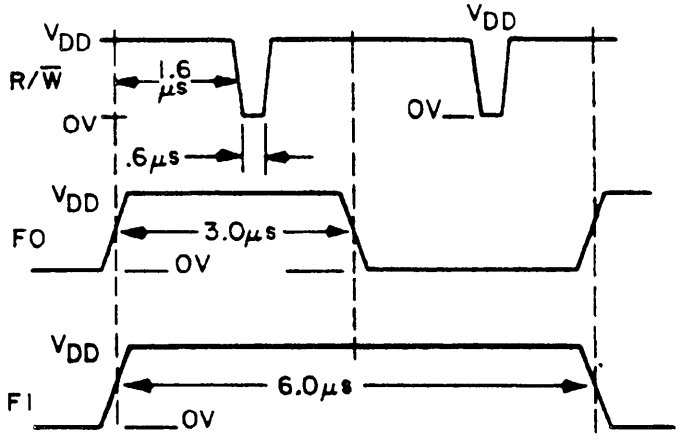
NOTES:
1. R = 2 kΩ.
2. V_{DD} = 11 V, V_{SS} = 0 V.
3. $V_{IH} = V_{DD}$, $V_{IL} = V_{SS}$.
4. Waveform for F2 = 1/2 F1, F3 = 1/2 F2, F4 = 1/2 F3 . . . . . F11 = 1/2 F10.

FIGURE 6. Dynamic burn-in and steady-state life-test circuit.

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Device type 02

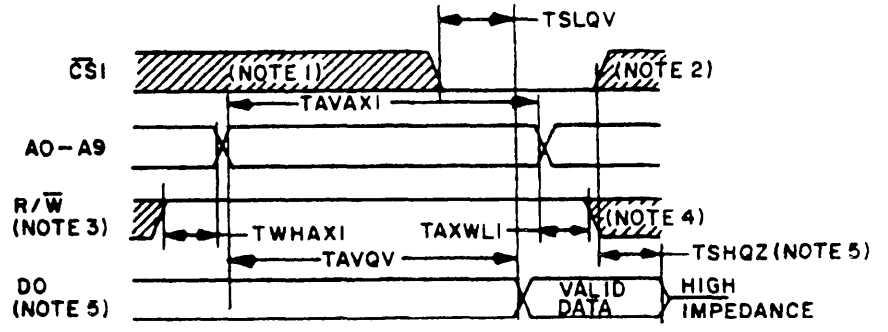




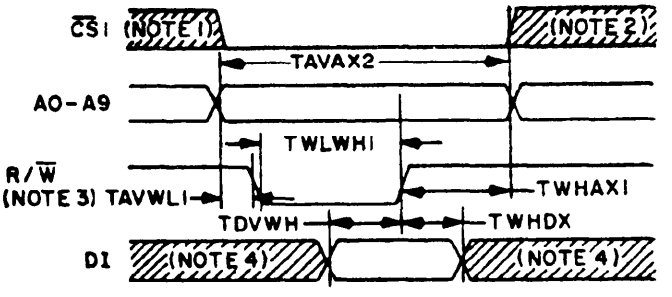
NOTES:
1. R = 5.6 kΩ.
2. V_{DD} = 11 V, V_{SS} = 0 V.
3. Waveforms for F2 = 1/2 F1, F3 = 1/2 F2, F4 = 1/2 F3 . . . . . F11 = 1/2 F10.

FIGURE 6. Dynamic burn-in and steady-state life-test circuit - Continued.

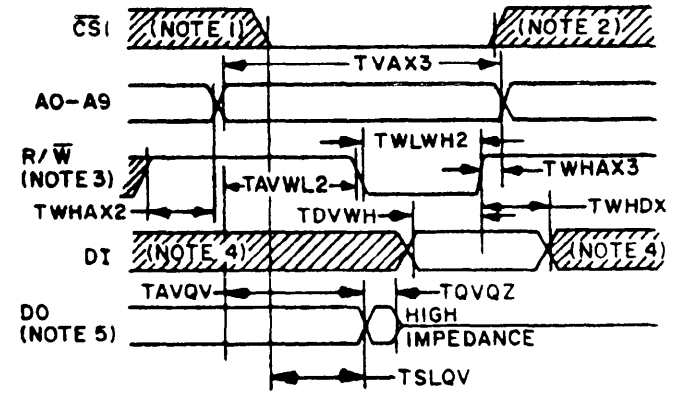
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READ-CYCLE-WAVEFORMS



WRITE-CYCLE WAVEFORMS



READ/MODIFY/WRITE-CYCLE WAVEFORMS

NOTES:
1. Chip-Select ($\overline{CS}$) permitted to change from high to low level or remain low on a selected device.
2. Chip-Select ($\overline{CS}$) permitted to change from low to high level or remain low.
3. Read/Write ($\overline{R/W}$) must be at a high level during all address transitions.
4. Don't care.
5. Data-Out (DO) is a high impedance within TEHQZ ns after the falling edge of $\overline{R/W}$ or the rising edge of $\overline{CS}$.
6. Timing parameter limits are as specified on page 16 using the circuit as shown on figure 9.

FIGURE 7. Read, write and read/modify/write waveforms for device type 01.

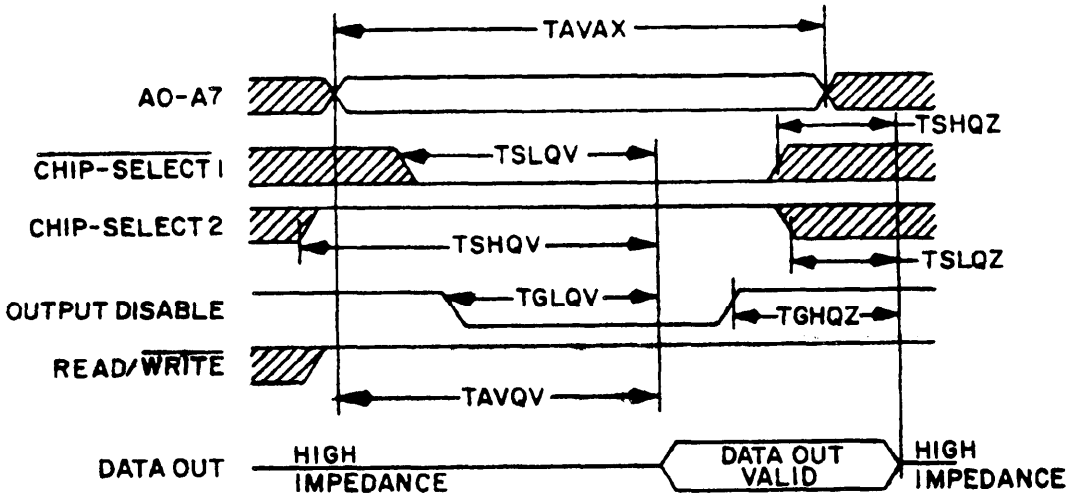
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Timing parameter	Description	Limits				Unit
		V_{DD} = 5 V		V_{DD} = 10 V		
		Min	Max	Min	Max	
TAVAX1	Read cycle time	360		230		ns
TAVQV	Read access time		260		140	ns
TSLQV	Output enable time	80		60		ns
TACWL	Read/write hold time	90		40		ns
TWHAX1	Read/write setup time	70		40		ns
TSHQZ	Output disable time		60		30	ns
TAVAX2	Write cycle time	360		230		ns
TWLWH	Write pulse width	150		90		ns
TAVWL1	Address setup time	70		40		ns
TWHAX2	Read/write setup time	90		40		ns
TDVWH	Data setup time	50		30		ns
TWHDX	Data hold time	50		30		ns
TAVAX3	R/M/W cycle time	550		330		ns
TAVWL2	Address setup time	250		125		ns
TQVQZ	Output disable time		60		30	ns
TWHAX3	Read/write setup time	60		40		ns

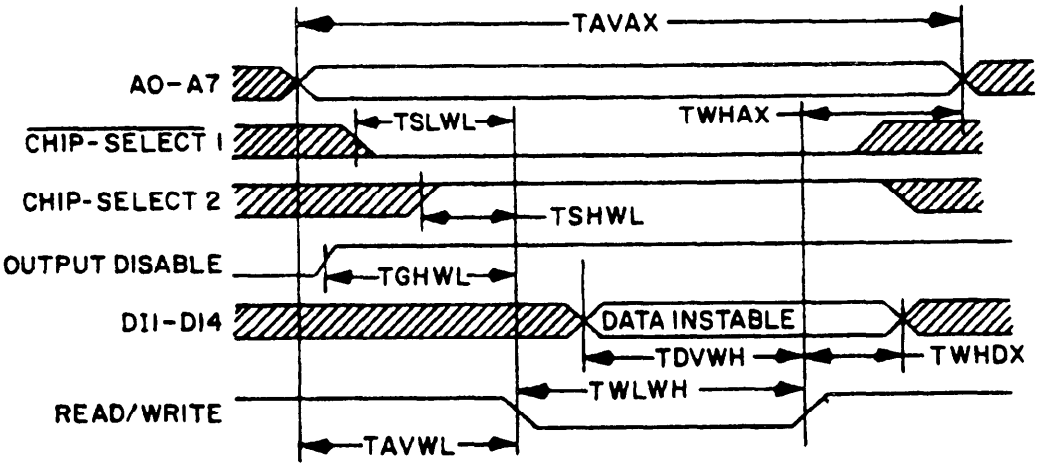
FIGURE 7. Read, write and read/modify/write waveforms for device type 01 - Continued.

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READ CYCLE WAVEFORMS AND TIMING DIAGRAM



WRITE CYCLE WAVEFORMS AND TIMING DIAGRAM

NOTE: Timing parameter limits are as specified on page 18 using the circuit as shown on figure 9.

FIGURE 8. Read and write waveforms for device type 02.

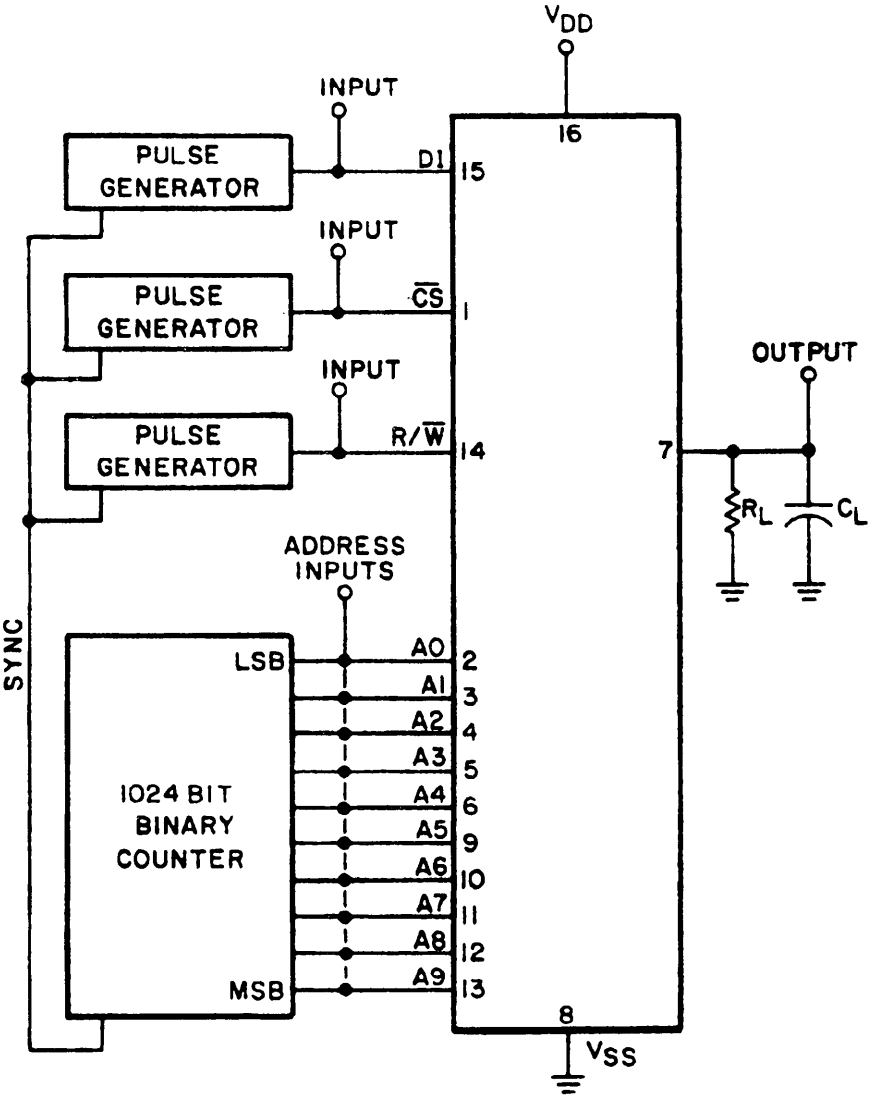
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Timing parameter	Description	Limits				Unit
		V_{DD} = 5 V		V_{DD} = 10 V		
		Min	Max	Min	Max	
TAVAX	Read or write cycle time	630		315		ns
TAVQV	Read access time		630		315	ns
TAVWL	Address setup time	220		100		ns
TWHAX	Read/write setup time		60		40	ns
TWLWH	Write pulse width		350		175	ns
TDVWH	Data setup time		290		150	ns
TWHDX	Data hold time		60		40	ns
TSLWL	Chip select ($\overline{CS1}$) setup time		50		30	ns
TSHWL	Chip select (CS2) setup time		50		30	ns
TGHWL	Output disable setup time		50		30	ns
TSLQV	Output valid from chip select 1 ($\overline{CS1}$)		560		280	ns
TSHQV	Output valid from chip select 2 (CS2)		560		280	ns
TGLQV	Output active from output disable		250		140	ns
TSHQZ	Output hold from chip select 1 ($\overline{CS1}$)	30		30		ns
TSLQZ	Output hold from chip select 2 (CS2)	30		30		ns
TGHQZ	Output hold from output disable	30		30		ns

FIGURE 8. Read and write waveforms for device type 02 - Continued.

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DEVICE TYPE 01

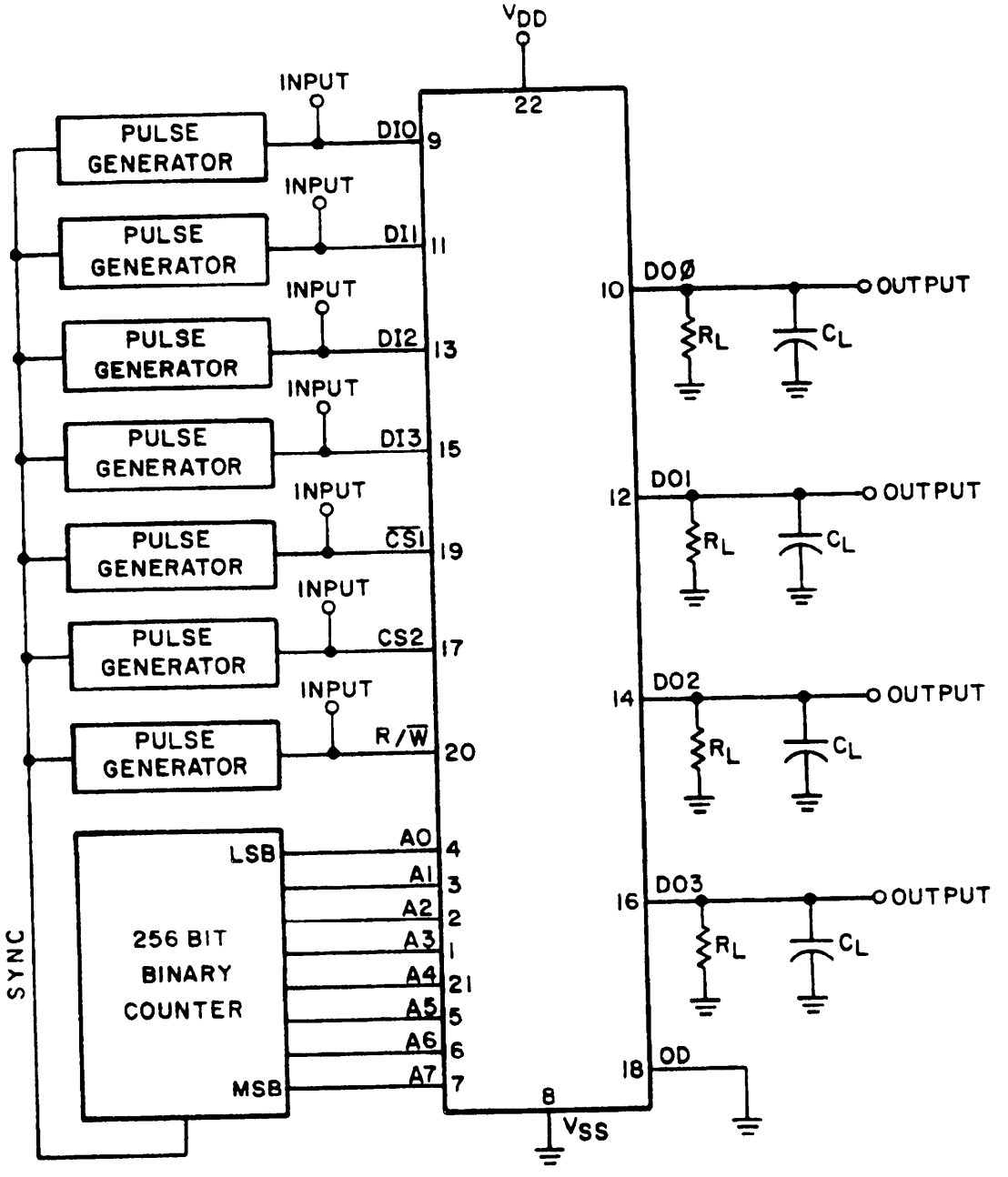


NOTES:
1. C_L = 50 pF ±10% including wiring and probe capacitance.
2. t_r and $t_f \leq 20$ ns.
3. All generator output impedance equals 50Ω.
4. All voltages measured with respect to V_{SS} terminal.
5. R_L = 2.7 kΩ.

FIGURE 9. Switching time test circuit.

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DEVICE TYPE 02

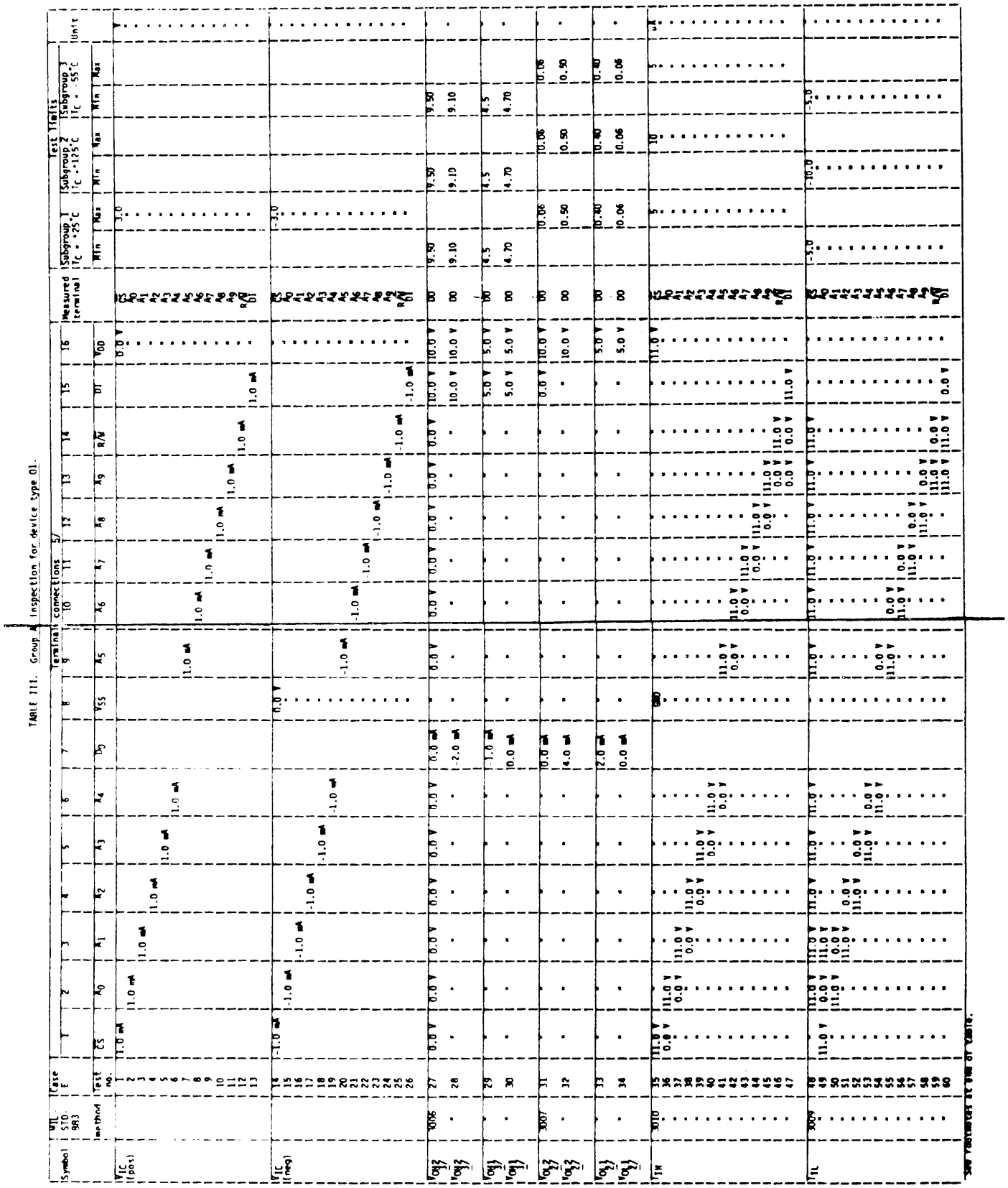


NOTES:
1. C_L = 50 pF ±10% including wiring and probe capacitance.
2. $t_r = t_f \leq 20$ ns.
3. All generator output impedance = 50Ω.
4. All voltages measured with respect to V_{SS} terminal.
5. R_L = 2.7 kΩ.

FIGURE 9. Switching time test circuit - Continued.

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TABLE III. Group A inspection for device type 01.



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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case f	Terminal connections 5/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Subgroups T_C =
		Test no.	$\overline{CS}$	A_0	A_1	A_2	A_3	A_4	D_0	V_{SS}	A_5	A_6	A_7	A_8	A_9	R/$\overline{W}$	DI	V_{DD}		Min
I_{ZH}		61	11.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	11.0 V	GND	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	11.0 V	DO	
I_{ZH}		62	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	11.0 V	"	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	"	"	
I_{ZL}		63	11.0 V	11.0 V	11.0 V	11.0 V	11.0 V	11.0 V	0.0 V	"	11.0 V	11.0 V	11.0 V	11.0 V	11.0 V	11.0 V	11.0 V	"	"	-10.
I_{ZL}		64	0.0 V	11.0 V	11.0 V	11.0 V	11.0 V	11.0 V	0.0 V	"	11.0 V	11.0 V	11.0 V	11.0 V	11.0 V	11.0 V	11.0 V	"	"	-10.
I_{DD1}/	3005	65	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V		"	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V		V_{DD}	
I_{DD2}/	"	66	"	"	"	"	"	"		"	"	"	"	"	"	"	"		"	
I_{DD3}/	"	67	"	"	"	"	"	"		"	"	"	"	"	"	"	"		"	
I_{DD4}/	"	68	"	"	"	"	"	"		"	"	"	"	"	"	"	"		"	
I_{SS1}/	"	69	"	11.0 V	11.0 V	11.0 V	11.0 V	11.0 V		"	11.0 V	11.0 V	11.0 V	11.0 V	11.0 V	11.0 V	11.0 V		V_{SS}	
I_{SS2}/	"	70	"	"	"	"	"	"		"	"	"	"	"	"	"	"		"	
I_{SS3}/	"	71	"	"	"	"	"	"		"	"	"	"	"	"	"	"		"	
I_{SS4}/	"	72	"	"	"	"	"	"		"	"	"	"	"	"	"	"		"	
I_{DDL}	"	73	"	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V		"	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	5.0 V	V_{DD}	
I_{DDOP}	"	74	12/	12/	12/	12/	12/	12/	12/	"	12/	12/	12/	12/	12/	12/	12/	10.0 V	V_{DD}	
																				Subgroups T_C =
																				Min
Functional tests 5/,6/,7/	3014	75	9/	9/	9/	9/	9/	9/	9/	GND	9/	9/	9/	9/	9/	9/	9/	9/	DO	9/
PWRDWN	3014	76	11/	11/	11/	11/	11/	11/	11/	GND	11/	11/	11/	11/	11/	11/	11/	11/	V_{DD}	
																				Subgroups T_C =
																				Min
TQVQZ		77	6/	6/	6/	6/	6/	6/	6/	GND	6/	6/	6/	6/	6/	6/	6/	5.0 V 10.0 V	DO	
TSLQV		78	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
TSHQZ		79	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
TAVQV		80	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases V	Terminal connections 5/																				Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	Measured terminal	Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
																										Min	Max	Min	Max	Min	Max	
I_{IH1}	3009	1	I_1	I_2	I_3	I_4	I_5	I_6	I_7	V_{SS}	O/I_6	I/O_6	DB_1	I/O_2	DB_3	DB_7	I/O_1	DB_5	CS_2	$R/\overline{W}$	CS_1	$R/\overline{W}$	I_4	V_{DD}	DB_0		5.0		20		5.0	μA
		2																							DB_1						
		3																							DB_2						
		4																							DB_3						
I_{IH2}	3010	5	I_1	I_2	I_3	I_4	I_5	I_6	I_7	V_{SS}	O/I_6	I/O_6	DB_1	I/O_2	DB_3	DB_7	I/O_1	DB_5	CS_2	$R/\overline{W}$	CS_1	$R/\overline{W}$	I_4	V_{DD}	DB_0		5.0		20		5.0	μA
		6																							DB_1						
		7																							DB_2						
		8																							DB_3						
I_{IH3}	3011	9	I_1	I_2	I_3	I_4	I_5	I_6	I_7	V_{SS}	O/I_6	I/O_6	DB_1	I/O_2	DB_3	DB_7	I/O_1	DB_5	CS_2	$R/\overline{W}$	CS_1	$R/\overline{W}$	I_4	V_{DD}	DB_0		5.0		20		5.0	μA
		10																							DB_1						
		11																							DB_2						
		12																							DB_3						
I_{IH4}	3012	13	I_1	I_2	I_3	I_4	I_5	I_6	I_7	V_{SS}	O/I_6	I/O_6	DB_1	I/O_2	DB_3	DB_7	I/O_1	DB_5	CS_2	$R/\overline{W}$	CS_1	$R/\overline{W}$	I_4	V_{DD}	DB_0		5.0		20		5.0	μA
		14																							DB_1						
		15																							DB_2						
		16																							DB_3						
I_{IH5}	3013	17	I_1	I_2	I_3	I_4	I_5	I_6	I_7	V_{SS}	O/I_6	I/O_6	DB_1	I/O_2	DB_3	DB_7	I/O_1	DB_5	CS_2	$R/\overline{W}$	CS_1	$R/\overline{W}$	I_4	V_{DD}	DB_0		5.0		20		5.0	μA
		18																							DB_1						
		19																							DB_2						
		20																							DB_3						
I_{IH6}	3014	21	I_1	I_2	I_3	I_4	I_5	I_6	I_7	V_{SS}	O/I_6	I/O_6	DB_1	I/O_2	DB_3	DB_7	I/O_1	DB_5	CS_2	$R/\overline{W}$	CS_1	$R/\overline{W}$	I_4	V_{DD}	DB_0		5.0		20		5.0	μA
		22																							DB_1						
		23																							DB_2						
		24																							DB_3						
I_{IH7}	3015	25	I_1	I_2	I_3	I_4	I_5	I_6	I_7	V_{SS}	O/I_6	I/O_6	DB_1	I/O_2	DB_3	DB_7	I/O_1	DB_5	CS_2	$R/\overline{W}$	CS_1	$R/\overline{W}$	I_4	V_{DD}	DB_0		5.0		20		5.0	μA
		26																							DB_1						
		27																							DB_2						
		28																							DB_3						
I_{IH8}	3016	29	I_1	I_2	I_3	I_4	I_5	I_6	I_7	V_{SS}	O/I_6	I/O_6	DB_1	I/O_2	DB_3	DB_7	I/O_1	DB_5	CS_2	$R/\overline{W}$	CS_1	$R/\overline{W}$	I_4	V_{DD}	DB_0		5.0		20		5.0	μA
		30																							DB_1						
		31																							DB_2						
		32																							DB_3						
I_{IH9}	3017	33	I_1	I_2	I_3	I_4	I_5	I_6	I_7	V_{SS}	O/I_6	I/O_6	DB_1	I/O_2	DB_3	DB_7	I/O_1	DB_5	CS_2	$R/\overline{W}$	CS_1	$R/\overline{W}$	I_4	V_{DD}	DB_0		5.0		20		5.0	μA
		34																							DB_1						
		35																							DB_2						
		36																							DB_3						
I_{IH10}	3018	37	I_1	I_2	I_3	I_4	I_5	I_6	I_7	V_{SS}	O/I_6	I/O_6	DB_1	I/O_2	DB_3	DB_7	I/O_1	DB_5	CS_2	$R/\overline{W}$	CS_1	$R/\overline{W}$	I_4	V_{DD}	DB_0		5.0		20		5.0	μA
		38																							DB_1						
		39																							DB_2						
		40																							DB_3						
I_{OZL}	3007	41	I_1	I_2	I_3	I_4	I_5	I_6	I_7	V_{SS}	O/I_6	I/O_6	DB_1	I/O_2	DB_3	DB_7	I/O_1	DB_5	CS_2	$R/\overline{W}$	CS_1	$R/\overline{W}$	I_4	V_{DD}	DB_0		5.0		20		5.0	μA
		42																							DB_1						
		43																							DB_2						
		44																							DB_3						
I_{OZH}	3007	45	I_1	I_2	I_3	I_4	I_5	I_6	I_7	V_{SS}	O/I_6	I/O_6	DB_1	I/O_2	DB_3	DB_7	I/O_1	DB_5	CS_2	$R/\overline{W}$	CS_1	$R/\overline{W}$	I_4	V_{DD}	DB_0		5.0		20		5.0	μA
		46																							DB_1						
		47																							DB_2						
		48																							DB_3						

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case outlines	Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	Measured terminal	Test limits Subgroups 5 T_C = +125°C		Subgroups 6 T_C = -55°C		Unit
				I_{13}	A_2	A_1	A_0	A_5	A_6	A_7	V_{SS}	M_0	DQ_0	DI_1	DO_1	DI_2	DO_2	DI_3	DO_3	CS_2	$\overline{OE}$	CS_1	$\overline{R/W}$	A_4	V_{DD}	V_{DD}		Min	Max	Min	Max	
				Subgroup 7 T_C = +25°C																											
I_{DDOP}			114	107	107	107	107	107	107	107		107	107	107	107	107	107	107	107	107	107	107	107	107	10.0 V	V_{DD}					mA
I_{DDSB1} 6/ I_{DDSB1} 6/			115																							00					
			116																												
I_{DDSB2} I_{DDSB2} 6/			117																												
I_{DDSB3} I_{DDSB3} 7/			118																												
I_{DDSB4}			119																							00					
				Subgroups 7 and 8 T_C = +25°C and +125°C																											
I_{DDOV}			120																								5.5 V 10 V		500 200		500 200	μA
I_{DDOV1}			121																									50		50	
I_{DDOV2}			122																										630 315		630 315	
I_{DDOV3}			123																										250 140		250 140	
I_{DDOV4}			124																									50		50	
I_{DDOV5}			125																												
I_{DDOV6}			126																									50		50	

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 - Continued.

1/ Turn off power to chip for ≥ 100 ms to be sure RANDOM data in device prior to test.

2/ Write a "0" into every cell.

3/ Write a "1" into every cell.

4/ Write checkerboard into memory.

5/ Pins not designated are open.

6/ The GALPAT pattern is as follows:

a. Write a "0" into all address locations.
b. Write a "1" into location i, read i+1, i, i+2, i, etc., where i goes through all address locations.
c. Repeat steps a and b with opposite data.

7/

V_{DD}	t_{cycle}	V_{IH}	V_{IL}
10 V	315 ns	8 V	2 V
5 V	630 ns	4 V	1 V

8/

T_C = +25°C		+125°C	-55°C	Unit
A	-1.2	-1.0	-1.2	mA
B	-2.5	-2.0	-2.5	
C	2.5	2.0	2.5	
D	5.0	4.0	5.0	

9/ Terminal conditions shall be as follows:

a. Inputs: High = 7.0 to 10.0 V, low = 0.0 to 3.0 V.
b. Outputs: High = 9.1 V minimum, low = 0.5 V maximum.
c. Functional tests shall be performed with V_{DD} = 10.0 V.

10/ Operating current test (I_{DDOP}) terminal conditions shall be as follows:

a. Use timing parameters and limits specified on figures 7 and 8.
b. Use topologically true checkerboard pattern (see figures 4 and 5 and footnotes 6 and 9.
c. Use load specified in figure 9.
d. V_{IL} = 0 V; V_{IH} = 5.5 V.

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TABLE III. Group A inspection for device type 02 - Continued.

11/ Power-down test (PWRDWN) terminal conditions shall be as follows:

a. Use timing parameters and limits specified on figures 7 and 8.
b. $V_{OL} \leq 0.45$ V; $V_{OH} \geq 2.4$ V.
c. Use load specified on figure 9.
d. $V_{IL} = 0$ V; $V_{IH} = 4.5$ V.
e. The power down test shall be performed as follows:
   (1) Power up RAM to $V_{DD} = 4.5$ volts.
   (2) Write topologically true checkerboard pattern into RAM memory array (see figures 4 and 5 and footnotes 6 and 9).
   (3) Reduce V_{DD} to 2.0 volts. All device inputs shall be reduced along with V_{DD}, exercising special care to avoid latch-up (See note in 4.5.)
   (4) Maintain RAM at reduced voltage for 500 ms (0.5 seconds) minimum.
   (5) Raise V_{DD} to 4.5 volts.
   (6) Verify RAM contents to be unchanged from pattern written in (2) above.
   (7) Repeat steps (1) through (6), lowering V_{DD} by 0.1 volt, until RAM contents do not verify. Read and record lowest V_{DD} value at which RAM contents verify. This value of V_{DD} shall not be greater than that specified in table III.
   (8) Repeat steps (1) through (7), except use the inverse of the first checkerboard pattern.

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4.5.2 Delta measurements. Delta measurements, as specified in table II, shall be made and recorded before and after the required burn-in screens and life tests to determine delta compliance. The electrical parameters to be measured, with associated delta limits are listed in table IV.

TABLE IV. Delta limits at 25°C.

Parameter	Delta limit
I_{SS}	Device type 01 41 µA
I_{SS}	Device type 02 71 µA
V_{OH1}	±10%
V_{OL1}	±10%
I_{ZL}	±10%
I_{ZH}	±10%

4.5.3 High voltage (V_{ZAP}) test of input protection circuits. All input terminals (up to a maximum of four) of the devices under test (DUT) shall be subjected to a voltage pulse from a 100 pF source charged to 400 volts. This destructive test shall be conducted as follows using the test circuit on figure 10.

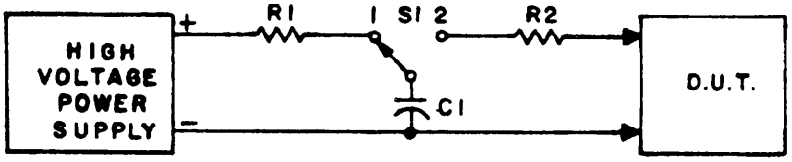
a. Measure I_{IH} and I_{IL} at the inputs selected, as stated above, at 25°C. The test limit for each input tested shall be ±1.0 µA at V_{DD} = 10 V dc. Measure I_{SS} on device under test (DUT) at +25°C. The test limit for this measurement shall be increased by a maximum of 20 percent of the specified I_{SS} table III limit at V_{DD} = 10.0 V dc.

b. V_{ZAP} is applied to the DUT in the following modes (see table V) by charging C1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high-voltage test.

Mode	+ Terminal	- Terminal
1	V_{DD}	Input
2	Input	V_{SS}
3	Input	Associated output

c. Within 24 hours repeat the I_{SS}, I_{IH}, and I_{IL} measurements on the same terminals as performed above. If a DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.



V_{ZAP} = 400 V minimum charge on C1.
10 MΩ ≤ R1 ≤ 50 MΩ.
R2 = 1.5 kΩ.
C1 = 100 pF.
S1 = Hg-wetted "bounceless" relay.

FIGURE 10. High voltage (V_{ZAP}) test circuit.

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4.5.4 Quiescent supply current (I_{SS}) test. When performing quiescent supply current measurements, the meter shall be placed so that all currents flow through the meter.

4.6 Data reporting. When specified in the contract or purchase order, a copy of the following data, as applicable, shall be supplied:

a. Attributes data for all screening tests (see 4.2) and variables data for all dynamic burn-in, and steady-state life tests.

b. The quality conformance inspection data (see 4.4).

c. Parameter distribution data on parameters evaluated during burn-in.

d. Final electrical parameters data.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510. The devices require electrostatic protection.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

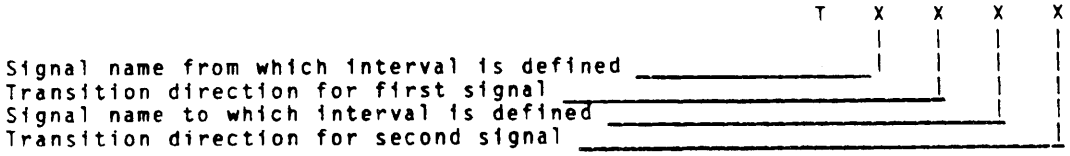
6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

V_{DD}	Supply voltage.
V_{SS}	Common or reference voltage node.
$\overline{A0}$ to A9	Address input.
$\overline{CS}$	Chip select input.
DI	Data input.
DO	Data output.
GND	Ground, zero voltage potential.
$\overline{R/W}$	Memory write input.
$\overline{OD}$	Memory read input (Output disable).
T_A	Free air temperature.
$V_{IC(pos)}$	Positive clampling input to V_{DD}.
$V_{IC(neg)}$	Negative clamping input to V_{SS}.

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I_Z	Output high-impedance leakage current.
ISS	Quiescent supply current.
TAVAC1	Read cycle time.
TAVQV	Read access time.
TSLQV	Output enable time.
TACWL	Read/Write hold time.
TWHAC1	Read/Write setup time.
TSHQZ	Output disable time.
TAVAC2	Write cycle time.
TWLWH	Write pulse width.
TAVWL	Address setup time.
TDVWH	Data setup time.
TWHDX	Data hold time.
TAVAC3	R/M/W cycle time.
TQVQZ	Output disable time.
TSLWL	Chip select setup time ($\overline{CS}1$).
TSHWL	Chip select setup time (CS2).
TGHWL	Output disable setup time.
TSLQV	Output valid from chip select ($\overline{CS}1$).
TSHQV	Output valid from chip select (CS2).
TGLQV	Output active from output disable.
TSHQZ	Output hold from chip select ($\overline{CS}1$).
TSLQZ	Output hold from chip select (CS2).
TGHQZ	Output hold from output disable.

6.4.1 Timing parameter abbreviations. All timing abbreviations use upper case characters with no subscripts. The initial character is always T and is followed by four descriptors. These characters specify two signal points arranged in a "from-to" sequence that define a timing interval. The two descriptors for each signal point specify the signal name and the signal transitions. Thus the format is:



a. Signal definitions:
   - A = Address
   - D = Data in
   - Q = Data out
   - W = Write enable
   - E = Chip enable
   - S = Chip select
   - G = Output enable

b. Transition definitions:
   - H = Transition to high
   - L = Transition to low
   - V = Transition to valid
   - X = Transition to invalid or don't care
   - Z = Transition to off (high impedance)
   - C = Transition change

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.6 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.4) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	1821
02	1822

6.8 Handling. CMOS microcircuits should be handled with certain precautions to avoid damage due to accumulation of static charge. These CMOS devices are fabricated with a silicon gate technology, including input protection, which reduces the susceptibility to damage. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment, tools, and personnel.

c. Do not handle devices by the leads.

d. Store devices in conductive carriers or conductive foam.

e. Avoid use of plastic, rubber, or silk in areas where CMOS devices are handled.

f. Maintain relative humidity above 50 percent.

6.9 Ordering guidance. Since the qualification and certification requirements have been removed from the specification, orders may be placed immediately.

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
NASA - NA

Review activity:
DLA - ES

Agent:
DLA - ES

Preparing activity:
NASA - NA

(Project 5962-K646-1)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/471A(NASA)	MICROCIRCUITS, DIGITAL, CMOS/SOS 1024 BIT RANDOM ACCESS MEMORY (RAM) MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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